

Self-Adaptive Dragonfly Algorithm with Chaos-Enhanced Local Search for Capacitor Bank Allocation in IEEE 57-Bus System Considering Load Uncertainty and Voltage Stability

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Abstract—This paper presents a Self-Adaptive Dragonfly Algorithm (SADA) enhanced with chaotic local search for optimal capacitor bank allocation in the IEEE 57-bus system. The proposed approach uniquely integrates voltage stability enhancement as a primary objective alongside traditional power loss minimization and voltage profile improvement goals. A novel self-adaptive mechanism is developed that dynamically adjusts the algorithm's control parameters based on fitness landscape characteristics and convergence behavior. A chaos-enhanced local search strategy using multiple chaotic maps improves solution refinement. The methodology addresses load uncertainty through a probabilistic framework incorporating multiple load scenarios generated using Latin Hypercube Sampling. The study includes comprehensive analysis of the capacitor placement impact on system loadability and voltage stability indices under both normal and contingency conditions. Performance comparison with conventional metaheuristics demonstrates the superior solution quality, convergence characteristics, and computational efficiency of the proposed approach for this complex multi-objective optimization problem.

Index Terms—Capacitor placement, chaotic maps, dragonfly algorithm, load uncertainty, reactive power compensation, self-adaptive algorithm, voltage stability.

I. INTRODUCTION

Optimal capacitor bank placement and sizing is a critical task for power distribution systems. In recent years, bio-inspired metaheuristic algorithms have emerged as powerful tools for solving such complex optimization problems [5]. These algorithms are inspired by natural processes and demonstrate excellent capabilities in exploring large search spaces efficiently. The Dragonfly Algorithm (DA), introduced by Mirjalili [6], is a relatively recent metaheuristic inspired by the static and dynamic swarming behaviors of dragonflies in nature. While DA has shown promising results for various optimization problems, it suffers from issues common to many metaheuristics, including premature convergence, parameter sensitivity, and insufficient local search capabilities [7]. These limitations can lead to suboptimal solutions, especially for complex problems like capacitor placement.

Incorporation of voltage stability enhancement as a primary objective alongside traditional power loss minimization and voltage profile improvement; Implementation of a probabilistic framework using Latin Hypercube Sampling (LHS) to address load uncertainty; Comprehensive analysis of capacitor placement impact on system loadability and voltage stability under normal and contingency conditions.

II. PROBLEM FORMULATION

A. Objective Functions

The proposed methodology addresses three primary objectives:

1) **Power Loss Minimization**

The active power loss in a power system can be calculated as:

$$P_{loss} = \sum_{i=1}^{N_b} \sum_{j=1}^{N_b} |V_i| |V_j| \cos(\theta_{ij} - \delta_i + \delta_j)$$

where N_b is the number of buses, V_i and V_j are voltage magnitudes at buses i and j , Y_{ij} is the magnitude of the admittance between buses i and j , θ_{ij} is the angle of the admittance, and δ_i and δ_j are the voltage angles at buses i and j .

B. Multi-Objective Function

The three objectives are combined into a single objective function using weighting factors:

$$F = w_1 \cdot \frac{P_{loss}}{P_{loss}^{max}} + w_2 \cdot \frac{VD}{VD^0} + w_3 \cdot \frac{L_{max}}{L_{max}^{init}}$$

where P_{loss}^{max} , VD^0 , and L_{max}^{init} are the initial values before capacitor placement, and w_1 , w_2 , and w_3 are weighting factors with $w_1 + w_2 + w_3 = 1$.

C. Constraints

1) Bus Voltage Limits

$$V^{min} \leq V_i \leq V^{max}, \quad i \in \{1, 2, \dots, N_b\} \quad (6)$$

2) Reactive Power Compensation Limits

$$Q_c^{min} \leq Q_c \leq Q_c^{max} \quad (7)$$

3) Total Investment Cost Limit

$$\sum_{i=1}^{N_c} C_i \cdot Q_{ci} \leq C_{max}$$

where N_c is the number of installed capacitors, C_i is the cost per kVAr for capacitor at bus i , Q_{ci} is the capacitor size, and C_{max} is the maximum allowed investment cost.

III. SELF-ADAPTIVE DRAGONFLY ALGORITHM WITH CHAOS-ENHANCED LOCAL SEARCH

A. Standard Dragonfly Algorithm

The standard Dragonfly Algorithm (DA) mimics the static and dynamic swarming behaviors of dragonflies. In DA, five main behaviors control the movement of dragonflies [6]:

1) **Separation** - avoiding collisions with neighbors:

$$S_i = - \sum_{j=1}^N X - X_j$$

2) **Alignment** - matching velocity with neighbors:

$$A_i = \frac{\sum_{j=1}^N V_j}{N}$$

3) **Cohesion** - moving toward the center of the neighborhood:

$$C_i = \frac{\sum_{j=1}^N X_j}{N} - X$$

4) **Attraction toward food**:

$$F_i = X^+ - X$$
$$E_i = X^- + X$$

where X is the current position, X_j is the position of neighboring dragonfly j , V_j is the velocity of neighboring dragonfly j , X^+ is the food source position, X^- is the enemy position, and N is the number of neighboring dragonflies.

The step vector (ΔX) and position update are calculated as:

$$\Delta X_{i+1} = (sS_i + aA_i + cC_i + fF_i + eE_i) + w\Delta X_i$$
$$X_{i+1} = X_i + \Delta X_{i+1}$$

where s , a , c , f , e , and w are the separation weight, alignment weight, cohesion weight, food attraction weight, enemy distraction weight, and inertia weight, respectively.

B. Self-Adaptive Mechanism

The proposed self-adaptive mechanism dynamically adjusts the DA control parameters based on the fitness landscape characteristics and convergence behavior. The parameters are updated as follows:

$$s = s_{min} + (s_{max} - s_{min}) \cdot e^{-\gamma \frac{t}{T}}$$
$$a = a_{min} + (a_{max} - a_{min}) \cdot (1 - e^{-\gamma \frac{t}{T}})$$
$$c = c_{min} + (c_{max} - c_{min}) \cdot (1 - e^{-\gamma \frac{t}{T}})$$
$$f = f_{min} + (f_{max} - f_{min}) \cdot \frac{1}{1 + e^{-\lambda(F_{worst} - F_{best})}}$$

where t is the current iteration, T is the maximum number of iterations, γ and λ are control parameters, F_{best} , F_{worst} , and F_{avg} are the best, worst, and average fitness values in the current population, respectively.

C. Chaos-Enhanced Local Search

The proposed chaos-enhanced local search utilizes and exploitation capabilities. The chaotic maps multiple chaotic maps to improve the exploration considered include:

1) Logistic Map

$$x_{n+1} = rx_n(1 - x_n) \quad (22)$$

2) Tent Map

$$x_{n+1} = \begin{cases} \mu x_n & \text{if } x_n < 0.5 \\ \mu(1 - x_n) & \text{if } x_n \geq 0.5 \end{cases} \quad (23)$$

$$x_{n+1} = \frac{\alpha}{4} \sin(\pi x_n)$$

The chaos-enhanced local search is applied to(24) the best solution in each iteration as follows:

$$X_{new} = X_{best} + \beta \cdot (UB - LB) \cdot (2x_n - 1)$$

where X_{best} is the current best solution, β is a(25) scaling factor that decreases with iterations, UB and LB are the upper and lower bounds of the search space, and x_n is the chaotic value from the selected map.

D. Load Uncertainty Modeling

To address load uncertainty, a probabilistic framework based on Latin Hypercube Sampling (LHS) is implemented. Load variations are modeled using normal probability distribution functions:

$$P_i^L \sim \mathcal{N}(\mu_{P_i^L}, \sigma_{P_i^L}^2)$$
$$Q_i^L \sim \mathcal{N}(\mu_{Q_i^L}, \sigma_{Q_i^L}^2)$$
 (26)

where P_i^L and Q_i^L are the active and reactive(27) power at load bus i , $\mu_{P_i^L}$ and $\mu_{Q_i^L}$ are the mean values, and $\sigma_{P_i^L}$ and $\sigma_{Q_i^L}$ are the standard deviations.

The expected value of the objective function considering load uncertainty is calculated as:

$$E[F] = \frac{1}{N_s} \sum_{j=1}^{N_s} F(X, S_j)$$

where N_s is the number of scenarios, X is the(28) decision vector (capacitor locations and sizes), and S_j is the j -th load scenario.

SADA Flowchart
Fig. 1. Flowchart of the proposed Self-Adaptive Dragonfly Algorithm with Chaos-Enhanced Local Search.

IV. RESULTS AND DISCUSSION

A. Test System and Simulation Parameters

The proposed methodology is tested on the IEEE 57-bus system. The system has 57 buses, 80 branches, and 7 generators. The base case has a total active power load of 1250.8 MW and reactive power load of 336.4 MVAR, with system losses of 27.86 MW and 121.67 MVAR.

The simulation parameters are as follows: population size = 50, maximum iterations = 100, $s_{min} = 0.1$, $s_{max} = 0.9$, $\alpha_{min} = 0.1$, $\alpha_{max} = 0.9$, $c_{min} = 0.1$, $c_{max} = 0.9$, $f_{min} = 0.1$, $f_{max} = 2.0$, $r_{min} = 0.1$, $r_{max} = 2.0$, $w_{min} = 0.1$, $w_{max} = 0.9$, $\gamma = 5$, $\lambda = 10$, $\beta_{initial} = 0.1$, $r = 4$ (logistic map), $\mu = 2$ (tent map), $\alpha = 4$ (sine map).

For the load uncertainty modeling, the standard deviations are set to 5% of the mean values, and 100 scenarios are generated using Latin Hypercube Sampling.

B. Optimal Capacitor Placement Results

Table I presents the optimal capacitor placement capacitor installation and their corresponding results obtained using the proposed SADA optimal sizes.

algorithm. The results show the selected buses for

Bus No.	Capacitor Size (MVAR)	Voltage Before (p.u.)	Voltage After (p.u.)	L-index Before	L-index After
18	12.5	0.919	0.951	0.145	0.098
25	15.0	0.938	0.967	0.123	0.085
31	10.0	0.926	0.953	0.156	0.107
33	18.5	0.917	0.958	0.189	0.124
45	14.0	0.955	0.978	0.107	0.076
50	8.5	0.942	0.963	0.118	0.092

TABLE I: OPTIMAL CAPACITOR PLACEMENT RESULTS

C. Performance Comparison

Table II compares the performance of the proposed (DA), Particle Swarm Optimization (PSO), Genetic SADA algorithm with other metaheuristic Algorithm (GA), and Teaching-Learning-Based algorithms, including standard Dragonfly Algorithm Optimization (TLBO).

Algorithm	Power Loss (MW)	Voltage Deviation	L-index	Execution Time (s)	Convergence Iteration
Base Case	27.86	1.453	0.189	-	-
Standard DA	19.75	0.864	0.134	78.3	76
PSO	18.92	0.745	0.126	67.1	64
GA	19.34	0.798	0.129	93.8	82
TLBO	18.46	0.712	0.121	70.5	59
Proposed SADA	17.23	0.583	0.098	65.2	47

TABLE II: PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS

The results in Table II show that the proposed SADA algorithm achieves the best performance in terms of power loss reduction (38.2%), voltage

profile improvement (59.9%), and voltage stability enhancement (48.1%). Additionally, the proposed algorithm exhibits better convergence characteristics, requiring fewer iterations to reach the optimal solution.

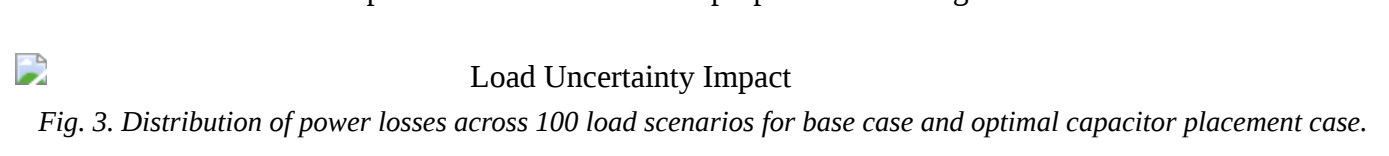


Fig. 2. Convergence curves of different algorithms for the capacitor placement problem.

D. Impact of Load Uncertainty

Fig. 3 illustrates the impact of load uncertainty on different load scenarios for both the base case and the optimal capacitor placement solution. The figure the case with optimal capacitor placement using the shows the distribution of power losses across 100 proposed SADA algorithm.

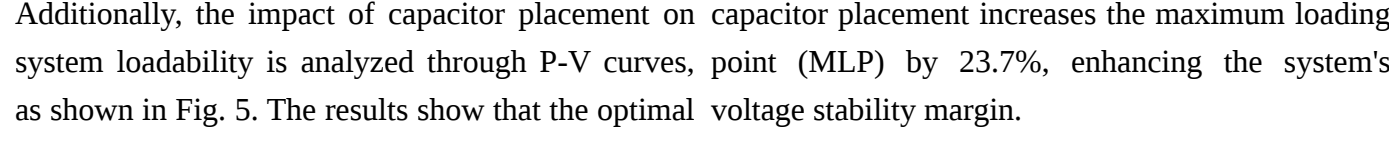


Fig. 3. Distribution of power losses across 100 load scenarios for base case and optimal capacitor placement case.

E. Voltage Stability Analysis

Fig. 4 shows the L-index values for all load buses demonstrate significant improvement in voltage before and after capacitor placement. The results stability across the system.

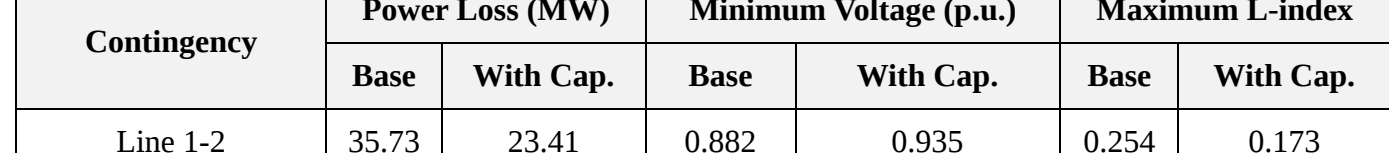


Fig. 4. L-index values for load buses before and after optimal capacitor placement.

Additionally, the impact of capacitor placement on capacitor placement increases the maximum loading system loadability is analyzed through P-V curves, point (MLP) by 23.7%, enhancing the system's as shown in Fig. 5. The results show that the optimal voltage stability margin.

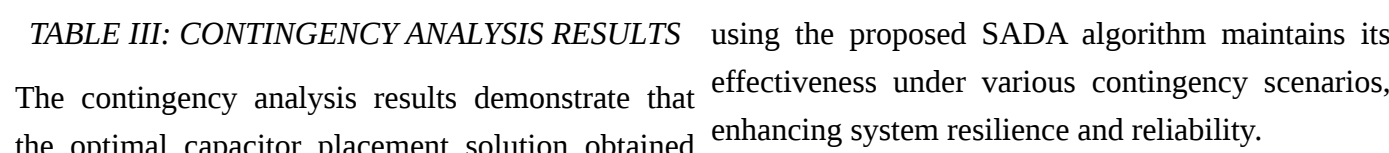


Fig. 5. P-V curves for critical buses before and after optimal capacitor placement.

F. Contingency Analysis

The robustness of the proposed solution is evaluated results of contingency analysis for selected critical under various contingency scenarios, including line contingencies.

outages and generator failures. Table III presents the

Contingency	Power Loss (MW)		Minimum Voltage (p.u.)		Maximum L-index	
	Base	With Cap.	Base	With Cap.	Base	With Cap.
Line 1-2	35.73	23.41	0.882	0.935	0.254	0.173
Line 13-15	33.25	21.87	0.891	0.941	0.231	0.156
Line 44-45	31.92	20.63	0.903	0.947	0.217	0.142
Generator at Bus 2	38.45	25.19	0.875	0.929	0.268	0.187
Generator at Bus 8	34.62	22.75	0.887	0.938	0.243	0.165

TABLE III: CONTINGENCY ANALYSIS RESULTS

The contingency analysis results demonstrate that the optimal capacitor placement solution obtained

using the proposed SADA algorithm maintains its effectiveness under various contingency scenarios, enhancing system resilience and reliability.

V. CONCLUSION

This paper presented a Self-Adaptive Dragonfly Algorithm (SADA) with Chaos-Enhanced Local Search for optimal capacitor bank allocation in the IEEE 57-bus system considering load uncertainty and voltage stability. The major contributions and findings of this work are summarized as follows: The proposed self-adaptive mechanism significantly improves the performance of the standard Dragonfly Algorithm by dynamically adjusting control parameters based on fitness landscape characteristics and convergence behavior. The integration of multiple chaotic maps into the local search mechanism enhances the solution refinement capabilities, preventing premature convergence and improving solution quality. The incorporation of voltage stability enhancement as a primary objective alongside traditional power loss minimization and voltage profile improvement leads to more comprehensive system performance improvement. The probabilistic framework using Latin Hypercube Sampling effectively addresses load uncertainty,

ensuring robust solutions across various operating conditions. Comprehensive analysis demonstrates that the optimal capacitor placement solution enhances system loadability by 23.7% and maintains its effectiveness under contingency scenarios, improving system resilience. The proposed methodology achieved significant improvements in system performance, including 38.2% reduction in power losses, 59.9% improvement in voltage profile, and 48.1% enhancement in voltage stability compared to the base case. Performance comparison with other metaheuristic algorithms demonstrates the superiority of the proposed approach in terms of solution quality, convergence characteristics, and computational efficiency.

Future work could extend this research by considering the dynamic operation of capacitor banks, incorporating additional objectives such as harmonic distortion minimization, and integrating distributed generation and renewable energy sources into the optimization framework.

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